Mixed Traffic Test (Serial)

2025-09-16-16-25-40

Test Setup Information

Overall Setup Info For all Tests	DUT Model	Test_DUT
	DUT Firmware	NA
	SSID	Test Configured
	Security	Test Configured
	No of Devices	2 (Virtual Clients: 0, Windows: 0, Linux: 1, Mac: 1, Android: 0 ,iOS: 0)
	Test Duration (HH:MM:SS)	00:03:00

Objective

The Candela mixed traffic test is designed to measure the access point performance andstability by running multiple traffic on real clients like Android, Linux, Windows, and IOSconnected to the access point. This test allows the user to choose multiple types of traffic likeclient capacity test, web browser test, video streaming test ping test. Along with theperformance measurements are client connection times, Station 4-Way Handshake time, DHCPtimes, and more. The expected behavior is for the AP to be able to handle all types of traffic onthe several stations (within the limitations of the AP specs) and Make sure all clients can run alltypes of traffic.

Traffic Details

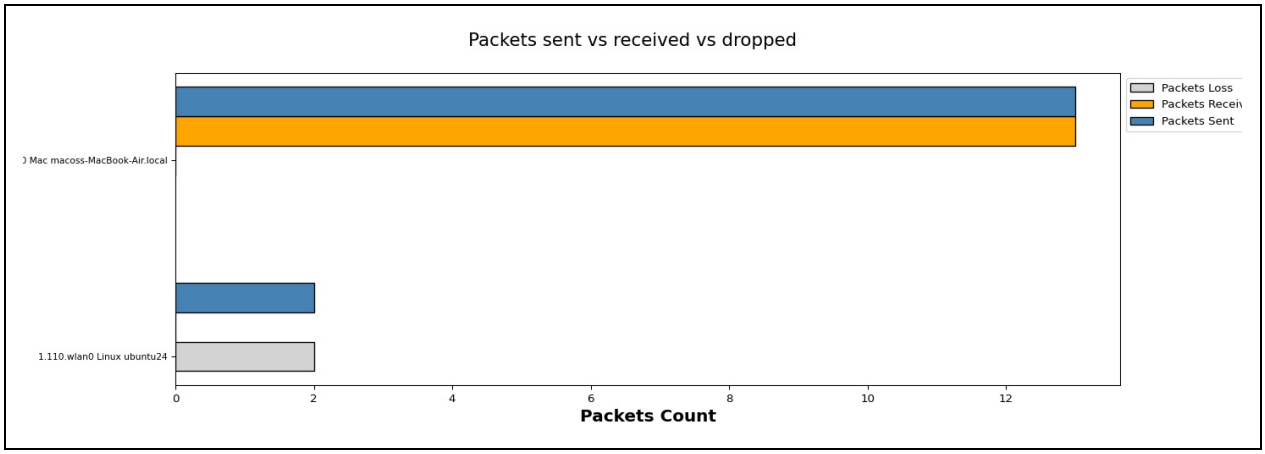
Sno	Test Cases	Test Duration	Test Status
1	Ping Test	1.0 minute	Executed
2	Quality Of Service(QOS) Test ['VI']	30 sec	Executed
3	FTP Test	30 sec	Executed
4	HTTP Test	30 sec	Executed
5	Multicast Test	30 sec	Executed

1. Ping Test

Test Configuration

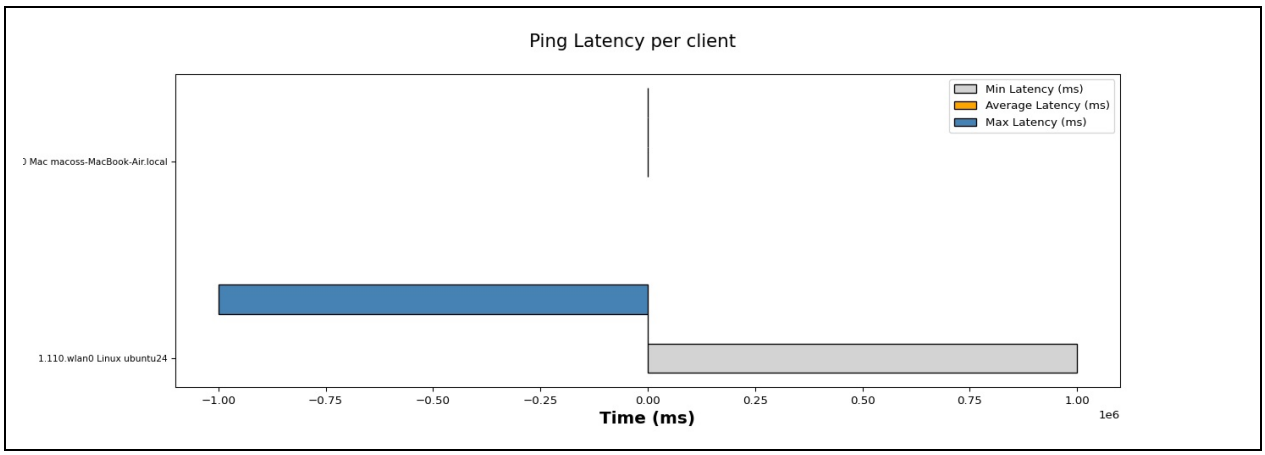
Test Setup Information	IP / Website	google.com

Packets sent vs packets received vs packets dropped



Wireless Client	MAC	Channel	SSID	Mode	Packets Sent	Packets Received	Packets Loss
ubuntu24 Linux	5c:5f:67:98:38:86	11	NETGEAR_2G_wpa2	802.11bgn 20 2x2	2	0	2
macoss-MacBook-Air.local Mac	30:35:ad:c5:e1:be	11	NETGEAR_2G_wpa2	802.11abgn-BE 20 2x2	13	13	0

Ping Latency Graph



Wireless Client	MAC	Channel	SSID	Mode	Min Latency (ms)	Average Latency (ms)	Max Latency (ms)
ubuntu24 Linux	5c:5f:67:98:38:86	11	NETGEAR_2G_wpa2	802.11bgn 20 2x2	1000000.000	0.000	-1000000.000
macoss-MacBook-Air.local Mac	30:35:ad:c5:e1:be	11	NETGEAR_2G_wpa2	802.11abgn-BE 20 2x2	19.668	24.826	42.256

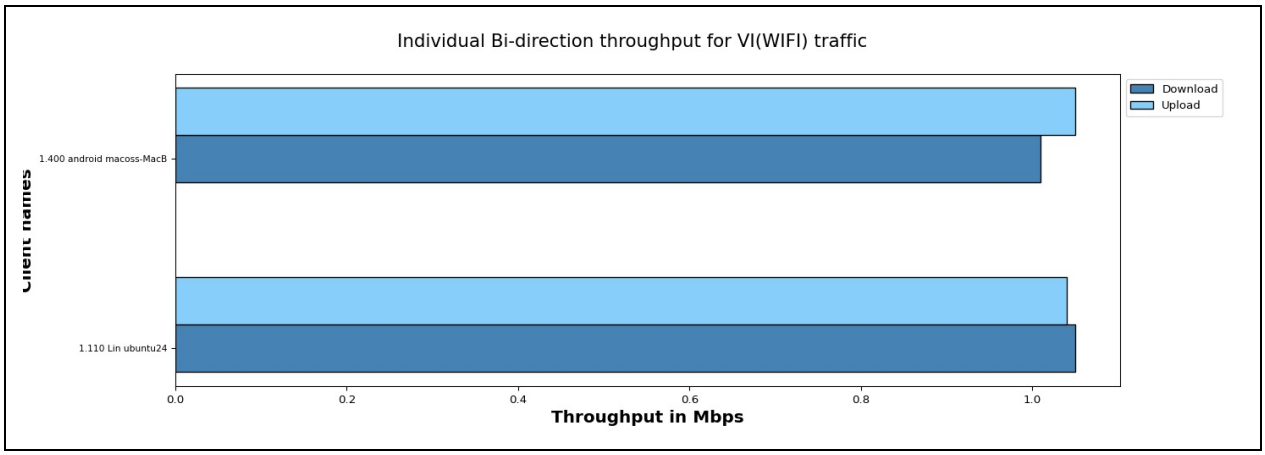
2. Quality Of Service(QOS) Test

Test Configuration

Test Setup Information	Protocol	TCP
	Traffic Direction	Bi-direction
	Security	Test Configured
	TOS	['VI']
	Per TOS Load	Upload:1.0 Mbps,Download:1.0 Mbps

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 2 clients running VI (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.110 Lin ubuntu24	5c:5f:67:98:38:86	NETGEAR_2G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.85 Mbps	0.84 Mbps	0.00	0.0
1.400 android macoss-MacB	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.69 Mbps	0.69 Mbps	0.08	0.0

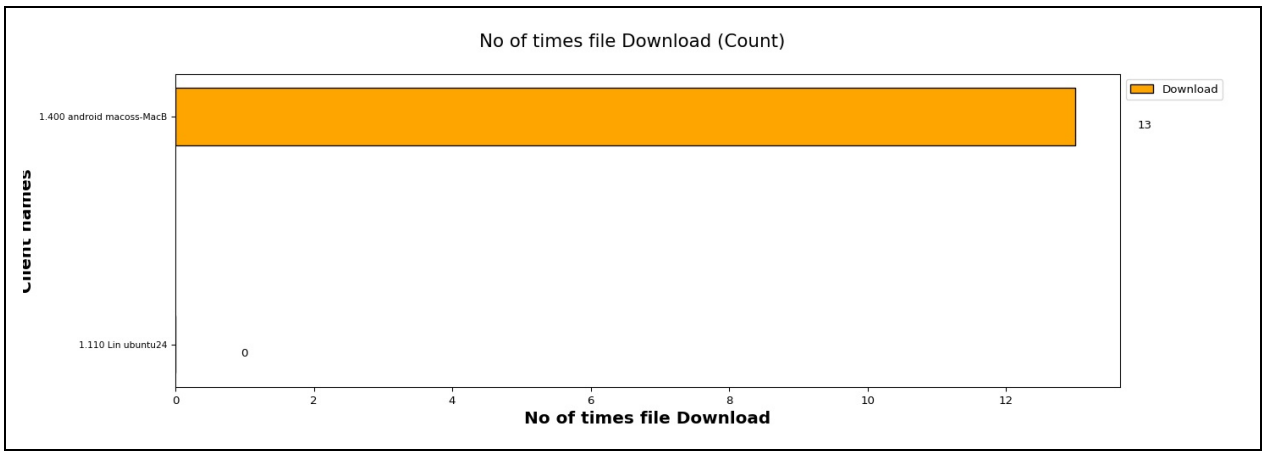
3. File Transfer Protocol (FTP) Test

Test Configuration

Test Setup Information	Traffic Direction	Download
	File Size	10MB
	File Location	/home/lanforge

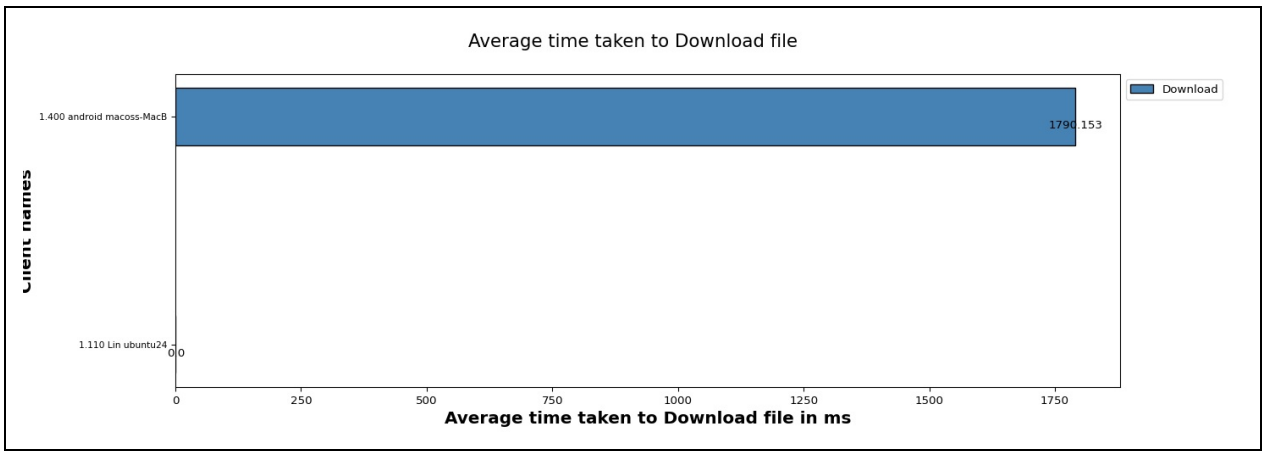
No.of times file Download

The below graph represents number of times a file Download for each client (WiFi) traffic. X-axis shows “No of times file Download” and Y-axis shows “Client names“.



Average time taken to Download file

The below graph represents average time taken to Download for each client (WiFi) traffic. X-axis shows "Average time taken to Download a file " and Y-axis shows "Client names".



Overall Results

Clients	MAC	Channel	SSID	Mode	No of times File downloaded	Time Taken to Download file (ms)	Bytes-rd (Mega Bytes)	RX RATE (Mbps)
1.110 Lin ubuntu24	5c:5f:67:98:38:86	11	NETGEAR_2G_wpa2	802.11bgn 20 2x2	0	0.000	0.000	0.0000
1.400 android macoss-MacB	30:35:ad:c5:e1:be	11	NETGEAR_2G_wpa2	802.11abgn-BE 20 2x2	13	1790.153	134.323	37.4697

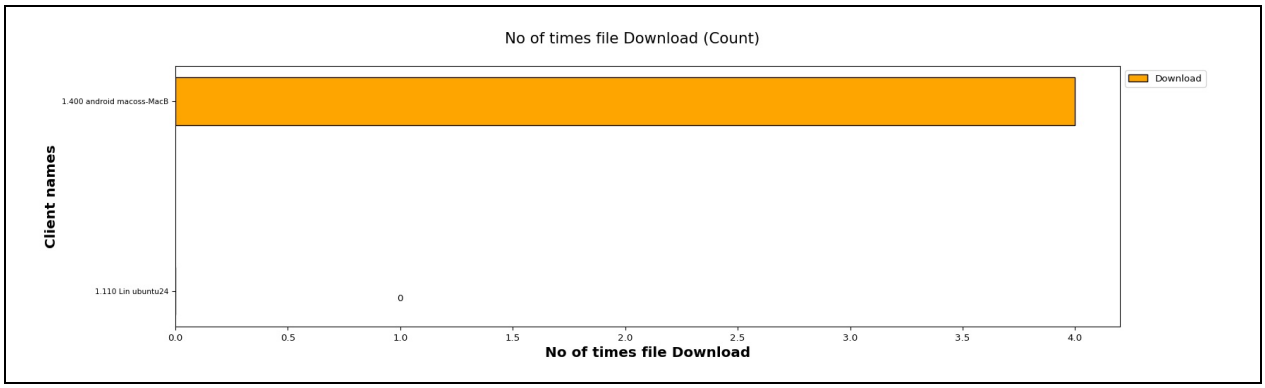
4. Hyper Text Transfer Protocol (HTTP) Test

Test Configuration

Test Setup Information	Traffic Direction	Download
	File Size	5MB
	File location	/usr/local/lanforge/nginx/html

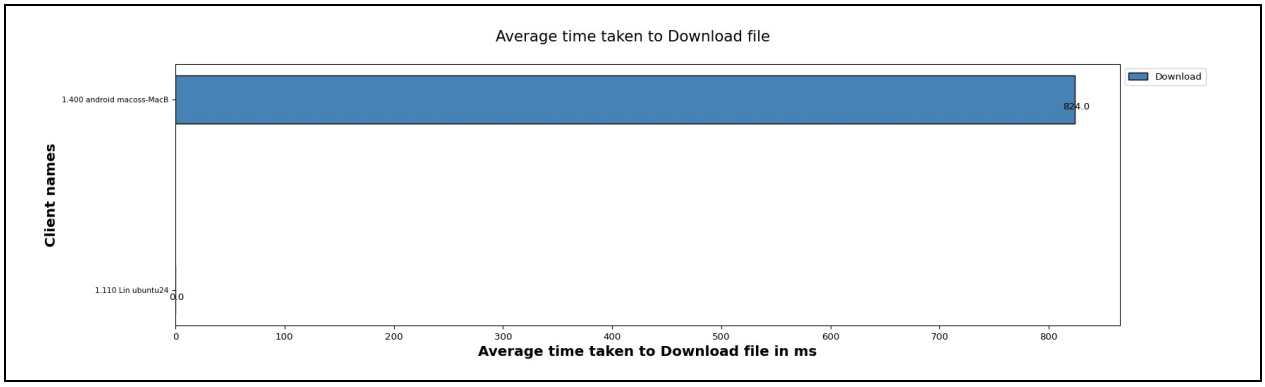
No of times file Downloads

The below graph represents number of times a file downloads for each client. X- axis shows "No of times file downloads and Y-axis shows Client names.



Average time taken to download file

The below graph represents average time taken to download for each client . X- axis shows "Average time taken to download a file " and Y-axis shows Client names.



Overall Results

Clients	MAC	Channel	SSID	Mode	No of times File downloaded	Average time taken to Download file (ms)	Bytes-rd (Mega Bytes)	Rx Rate (Mbps)
1.110 Lin ubuntu24	5c:5f:67:98:38:86	11	NETGEAR_2G_wpa2	802.11bgn 20 2x2	0	0.0	0.0	0.0000
1.400 android macoss-MacB	30:35:ad:c5:e1:be	11	NETGEAR_2G_wpa2	802.11abgn-BE 20 2x2	4	824.0	20.0	5.5675

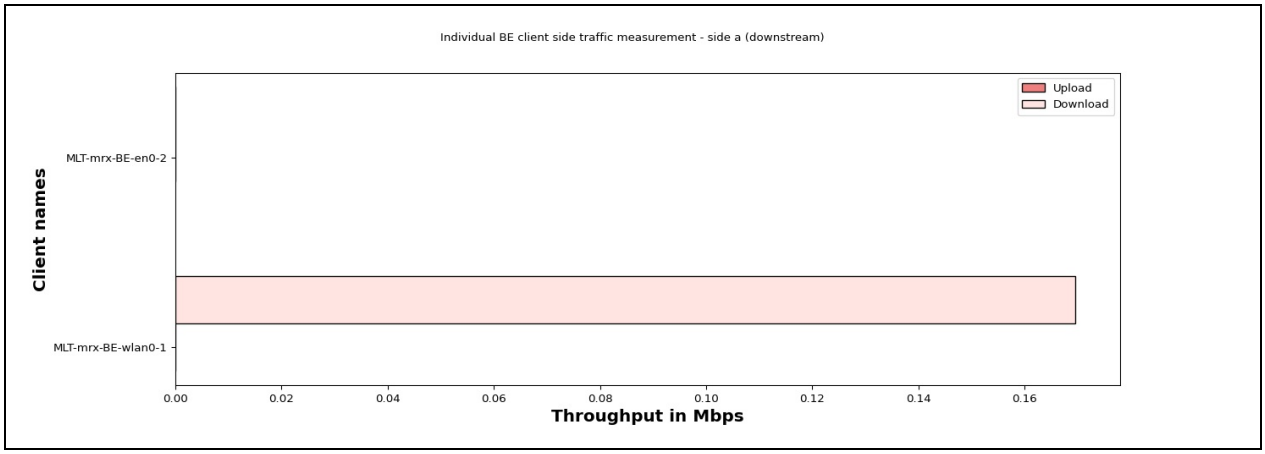
5. Multicast Test

Test Configuration

Test Setup Information	Protocol	UDP
	Type of Service (TOS)	BE
	Upstream Port	1.1.eth1
	Offered Load (Mbps)	3.0

Individual throughput with intended load 3.0 Mbps station for traffic BE (WiFi).

The below graph represents individual throughput for 2 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



Client Name	Endp Name	HW Version	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Download Rate Per Client (Mbps)	Download Rate Per Client (Mbps)	Download Drop Percentage (%)
1.110_ubuntu24_Linux	MLT-mrx-BE-wlan0-1	Linux/x86-64	1.110.wlan0	802.11bgn 20 2x2	5c:5f:67:98:38:86	NETGEAR_2G_wpa2	11	BE	Mcast	3.0	0.169496	0.0
1.400_macoss-MacBook-Air.local_Apple	MLT-mrx-BE-en0-2	Apple/x86-64	1.400.en0	802.11abgn-BE 20 2x2	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	11	BE	Mcast	3.0	0.000000	0.0

Overall Info	contact	support@candelatech.com